CIE 6002 Matrix Analysis Homework 1

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Declaration: I used LLM for enhancing language, explaining definitions and checking grammar in this homework.

1. Problem 1.2

(a) No. Suppose e and e' are both identity elements. Then

$$e = e \circ e' = e',$$

so the identity is unique.

(b) No, let $x \in g$. Suppose y and z are both inverses of x, then $y \circ x = e$ and $x \circ z = e$. Consider $y \circ x \circ z$:

$$y \circ (x \circ z) = y \circ e = y$$

$$(y \circ x) \circ z = e \circ z = z$$

By the associativity of the group operation, $y = z$.

(c) The inverse of the product is:

$$(x_1 \circ x_2 \circ \cdots \circ x_m)^{-1} = y_m \circ y_{m-1} \circ \cdots \circ y_1$$

Proof. We verify this by multiplying the proposed inverse by the original element:

$$\begin{aligned}
(x_1 \dots x_m) \circ (y_m \dots y_1) &= x_1 \dots (x_m \circ y_m) \dots y_1 \\
&= x_1 \dots (e) \dots y_1 \\
&= x_1 \dots x_{m-1} \circ y_{m-1} \dots y_1 \\
&\vdots \\
&= e
\end{aligned}$$

□

2. Problem 1.3

We will focus on the case $n = 2$. The result generalizes to $n > 2$ by embedding the 2×2 matrices into larger matrices. The logic for Hermitian matrices $H(n)$ is the same as that of $S(n)$ since real symmetric matrices are a subset of Hermitian matrices. The partial order $A \leq B$ is defined as:

$$B - A \succeq 0$$

meaning $B - A$ is positive semidefinite (PSD).

First, we show that the supremum of two specific projection matrices does not exist. Let A and B be:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Assume that the supremum $S = \sup(A, B)$ exists. Since the identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ satisfies $I - A \succeq 0$ and $I - B \succeq 0$, I is an upper bound for A and B. By the least upper bound definition of the supremum, we must have $S \leq I$. Let $S = \begin{pmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{pmatrix}$. Since $S \leq I$, the matrix $I - S$ must be PSD. This implies

$$1 - s_{11} \geq 0 \implies s_{11} \leq 1 \tag{1}$$

$$1 - s_{22} \geq 0 \implies s_{22} \leq 1 \tag{2}$$

Since S is an upper bound, $S \geq A$. Thus $S - A \succeq 0$:

$$\begin{pmatrix} s_{11} - 1 & s_{12} \\ s_{12} & s_{22} \end{pmatrix} \succeq 0$$

This requires $s_{11} - 1 \geq 0 \implies s_{11} \geq 1$. Combining with Eq. (1), we must have $s_{11} = 1$. Similarly, $S \geq B$ implies $s_{22} \geq 1$, and combined with Eq. (2), we get $s_{22} = 1$. Now check the determinant of $S - A$:

$$\det(S - A) = (s_{11} - 1)s_{22} - s_{12}^2 = -s_{12}^2$$

For $S - A$ to be PSD, the determinant must be non-negative:

$$-s_{12}^2 \geq 0 \implies s_{12} = 0$$

Therefore, if a supremum S exists, it must be the identity matrix I.

To disprove that I is the supremum, we need to find another upper bound C such that $I \nleq C$. If such a C exists, then I cannot be the least upper bound, which contradicts the result above. Consider the matrix $C = \begin{pmatrix} 2.5 & 2 \\ 2 & 3 \end{pmatrix}$, we check if C is an upper bound for A:

$$C - A = \begin{pmatrix} 1.5 & 2 \\ 2 & 3 \end{pmatrix}$$

Trace $= 4.5 > 0$. Determinant $= 0.5 > 0$. Thus $C \geq A$. Check if C is an upper bound for B:

$$C - B = \begin{pmatrix} 2.5 & 2 \\ 2 & 2 \end{pmatrix}$$

Trace = 4.5 > 0. Determinant = 1 > 0. Thus $C \geq B$. However,

$$C - I = \begin{pmatrix} 1.5 & 2 \\ 2 & 2 \end{pmatrix}$$

Determinant = −1 < 0. Since the determinant is negative, $C - I$ is not PSD. Therefore, $I \not\leq C$.

Contradiction: we showed that if $\sup(A, B)$ exists, it must be I. However, we found an upper bound C such that $I \not\leq C$. This means I fails the definition of a least upper bound. Therefore, $\sup(A, B)$ does not exist.

Since $S(n)$ for $n > 1$ contain $S(2)$ as a submatrix, this failure of the lattice property extends to all dimensions $n > 1$. This argument also applies to Hermitian matrices $H(n)$, as real symmetric matrices are a subset of Hermitian matrices.

3. Problem 1.4

Show that $(\mathbb{R}_n, \prec_w)$ is a lattice. For $x, y \in \mathbb{R}_n$, $x \prec_w y$ is defined as:

$$\sum_{i=1}^{k} x_i \leq \sum_{i=1}^{k} y_i, \quad k = 1, 2, \ldots, n$$

Let $S_k(x) = \sum_{i=1}^{k} x_i$ be the sum of the first k components of vector x. To compute the meet, we need to find a largest vector z such that $z \prec_w x$ and $z \prec_w y$. This means $S_k(z) \leq S_k(x)$ and $S_k(z) \leq S_k(y)$ for all k. The largest such $S_k(z)$ is:

$$S_k(z) = \min(S_k(x), S_k(y))$$

To recover z from $S_k(z)$, note that:

$$z_k = S_k(z) - S_{k-1}(z)$$

Thus, the meet $z = x \wedge y$ has components:

$$z_k = \min\left(\sum_{i=1}^{k} x_i, \sum_{i=1}^{k} y_i\right) - \min\left(\sum_{i=1}^{k-1} x_i, \sum_{i=1}^{k-1} y_i\right)$$

Similarly define

$$T_\vee(k) = \max\{S_x(k), S_y(k)\}, \qquad k = 0, 1, \ldots, n,$$

and let $\widehat{T}_\vee$ be the least concave majorant of $T_\vee$. Set

$$(x \vee y)_k^{\downarrow} := \widehat{T}_\vee(k) - \widehat{T}_\vee(k-1), \qquad k = 1, \ldots, n.$$

Then $x \vee y$ is the least upper bound of $\{x, y\}$ under $\prec_w$. The concavity condition enforces nonincreasing sorted components.

(a) The lattice is *not distributive* and also *not modular*. The operation of taking the least concave majorant in the join calculation destroys the distributive property. $x \wedge (y \vee z) \neq (x \wedge y) \vee (x \wedge z)$ in general cases.

A lattice is modular if $z \preceq x$ implies $x \wedge (y \vee z) = (x \wedge y) \vee z$. Similarly, this condition is destroyed by the least concave majorant operation.

(b) No.

For it to be a sublattice, the Meet and Join of any two integer vectors must result in an integer vector. The least concave majorant in the join operation creates a rational number but not necessarily an integer.

Consider $n = 3$ with integer vectors x and y such that their maximum partial sum sequence T contains $(1, 5)$ and $(3, 8)$, with $T_2 = 6$ to be skipped. The $(1, 5)$ will be connected directly to $(3, 8)$. The slope would be:

$$\frac{8-5}{3-1} = \frac{3}{2} = 1.5.$$

Since 1.5 is not an integer, $x \vee y \notin \mathbb{N}_n$. Thus, it is not a sublattice.

4. Problem 1.6

(a) $(P_n, \prec)$ is *not* a poset, because $\prec$ fails anti-symmetry on ordered vectors: if q is a nontrivial reordering of p, then $p^{\downarrow} = q^{\downarrow}$, hence $p \prec q$ and $q \prec p$, but $p \neq q$.

On the quotient $P_n/\simeq$, anti-symmetry holds: if $p \prec q$ and $q \prec p$, then $p^{\downarrow} = q^{\downarrow}$, so p and q have the same multiset of entries, i.e. $p = q$. Thus $(P_n/\simeq, \prec)$ *is* a poset.

(b) Yes, $(P_n/\simeq, \prec)$ is a lattice. Represent each class p by the sorted element $p^{\downarrow}$ (nonincreasing). For sorted $p, q \in P_n$, define partial sums

$$S_p(k) = \sum_{i=1}^{k} p_i, \qquad S_q(k) = \sum_{i=1}^{k} q_i, \qquad k = 0, 1, \ldots, n,$$

with $S_p(0) = S_q(0) = 0$ and $S_p(n) = S_q(n) = 1$.

Define

$$T_\wedge(k) = \min\{S_p(k), S_q(k)\}, \qquad T_\vee(k) = \max\{S_p(k), S_q(k)\}.$$

The partial sums of a sorted probability vector must be concave in k. Let $\widehat{T}_\wedge$ be the least concave majorant of $T_\wedge$ on $\{0, 1, \ldots, n\}$ (and similarly $\widehat{T}_\vee$ for $T_\vee$). Then define

$$(p \wedge q)_k := \widehat{T}_\wedge(k) - \widehat{T}_\wedge(k-1), \qquad (p \vee q)_k := \widehat{T}_\vee(k) - \widehat{T}_\vee(k-1),$$

for $k = 1, \ldots, n$. By construction, $p \wedge q$ is the greatest lower bound and $p \vee q$ is the least upper bound under $\prec$.

(c) Let $h(t) = -t \ln t$ on $[0, 1]$ with $h(0) = 0$. Then h is concave.

If $p \prec q$, w.l.o.g., we assume p and q are sorted nonincreasingly. Define

$$P_k := \sum_{i=1}^{k} p_i, \quad Q_k := \sum_{i=1}^{k} q_i$$

According to the definition of $\prec$, we have $P_k \leq Q_k$ for all k, and $P_n = Q_n$.

Next, we need to prove that if h is concave and $p \prec q$, then $\sum_{i=1}^{n} h(p_i) \geq \sum_{i=1}^{n} h(q_i)$. After that, we can show that H is Schur-concave.

Let $D = \sum_{i=1}^{n} h(q_i) - \sum_{i=1}^{n} h(p_i)$, so we need to show that $D \leq 0$.

Proof. Due to the concavity of h, we have by definition:

$$D = \sum_{i=1}^{n} h(q_i) - h(p_i) \leq \sum_{i=1}^{n} h'(p_i)(q_i - p_i). \tag{3}$$

Let $\Delta_k = Q_k - P_k$, then, we can rewrite Eq. (3) as:

$$
\begin{aligned}
D \leq \sum_{i=1}^{n} h'(p_i)(\Delta_i - \Delta_{i-1}) &= \sum_{i=1}^{n-1} \Delta_i (h'(p_i) - h'(p_{i+1})) \\
&= h'(p_1)(\Delta_1 - 0) \\
&+ h'(p_2)(\Delta_2 - \Delta_1) \\
&+ \dots \\
&+ h'(p_n)(\Delta_n - \Delta_{n-1}) \\
&= \Delta_1(h'(x_1) - h'(x_2)) + \Delta_2(h'(x_2) - h'(x_3)) \\
&+ \cdots + \Delta_{n-1}(h'(x_{n-1}) - h'(x_n)) + \underbrace{\Delta_n}_{=0} h'(x_n) \\
&= \sum_{i=1}^{n-1} \underbrace{\Delta_i}_{\geq 0} \underbrace{(h'(p_i) - h'(p_{i+1}))}_{\leq 0} \\
&\leq 0
\end{aligned}
$$

□

Hence, we have shown that if $p \prec q$, then $\sum_{i=1}^{n} h(p_i) \geq \sum_{i=1}^{n} h(q_i)$. Therefore, the entropy H is Schur-concave.

5. Problem 1.10

Let $u_i = \log x_i$, $v_i = \log y_i$ for $i = 1, \ldots, n$. Then $x \prec_{\log} y$ is equivalent to $u \prec v$. w.l.o.g, we assume x and y are sorted in nonincreasing order. In order to show that $x \prec_w y$, we need to prove that for all $k = 1, \ldots, n$,

$$\sum_{i=1}^{k} x_i \leq \sum_{i=1}^{k} y_i.$$

This is equivalent to showing that for all k,

$$\sum_{i=1}^{k} e^{u_i} \leq \sum_{i=1}^{k} e^{v_i}.$$

Similar as proved in Problem 1.6 (c), if a function f is convex, then $p \prec q$ implies $\sum_{i=1}^{n} f(p_i) \leq \sum_{i=1}^{n} f(q_i)$. Since e^t is convex, let $F_k(x) = \sum_{i=1}^{k} e^{x_i}$, we have $u \prec v$ implies $F_k(u) \leq F_k(v)$ for all k. Therefore,

$$\sum_{i=1}^{k} x_i \leq \sum_{i=1}^{k} y_i$$

for all $k = 1, \ldots, n$. This shows that $x \prec_w y$.